

MARIA: Towards Efficient Approximate Maximum Inner Product Search on Large-Scale Vector Datasets

Abstract

As the cardinality and diversity of vector datasets increase, driven by the success of numerous embedding models for unstructured data, the approximate maximum inner product search (AMIPS) problem in high-dimensional vector spaces becomes increasingly important for various machine learning tasks. Existing AMIPS methods, such as locality-sensitive hashing (LSH) and graph-based methods, suffer from inadequate query accuracy, low query efficiency, and limited scalability for large-scale datasets.

To address these challenges, we propose a novel graph-based framework called MARIA for efficiently solving the AMIPS problem. Specifically, MARIA first partitions a dataset into different blocks based on norms of points and constructs an approximate proximity graph (APG) for each block to capture inner product relationships, thereby alleviating norm bias issues. It then connects all APGs by linking points across different APGs based on their angular similarity, facilitating fast AMIPS queries on inter-block APGs. We also introduce a lightweight LSH index to mitigate the significant cost of constructing the novel APG from scratch. To further enhance query efficiency and scalability, we develop a skip-list-like strategy to reduce the overall size of the APG, along with an LSH-based quantization technique to lower query costs. Additionally, we provide a theoretical analysis of query quality and computational complexity. Extensive experiments on real-world datasets demonstrate that MARIA outperforms existing methods in terms of query efficiency, accuracy, and scalability for large-scale datasets.

Keywords

Maximum inner product search, Graph based method

ACM Reference Format:

. 2025. MARIA: Towards Efficient Approximate Maximum Inner Product Search on Large-Scale Vector Datasets . In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference SIGMOD '25)*. ACM, New York, NY, USA, 14 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

In recent years, there has been a significant surge in the availability of high-dimensional vector data (*i.e.*, vector representations of objects), primarily due to the proliferation of unstructured data and the remarkable success of deep neural network-based embedding models. These models have demonstrated their ability to embed various

types of unstructured data, including videos [13], molecular structural information of drugs [9], and documents [30], into vectors. Subsequently, these vectors can be stored in a vector database to facilitate a wide range of applications, including retrieval-augmented generation (RAG) in large language models (LLMs), which employ vector search to mitigate LLM hallucinations. As a result, the *Maximum Inner Product Search (MIPS) problem*, which aims to find a vector (*i.e.*, a representation of an object) from a dataset D that has the maximum inner product with a given query q , plays a crucial role in these applications. Given a query point $q \in \mathbb{R}^d$ and a dataset $\mathcal{D} \subseteq \mathbb{R}^d$, the MIPS problem aims to find a point¹ x in $\mathcal{D}$ that has the maximum inner product $q^\top x$. A basic solution to the MIPS problem is the classic tree-based method [26, 28, 40], which works well in low-dimensional spaces. However, when the dimensionality increases to several hundred or higher, the search time increases exponentially [2, 7, 10] due to the “curse of dimensionality.” Therefore, a more efficient and sufficient alternative is to find the approximate MIPS result. In this paper, we study the problem of *Approximate Maximum Inner Product Search (AMIPS)* (see Definition 2), which identifies the top- k “approximately” most relevant objects for query q over the collection $\mathcal{D}$.

Existing solutions for the AMIPS problem mainly consist of locality-sensitive hashing (LSH)-based methods [3, 21, 22] and graph-based methods [20, 23, 34]. LSH-based methods first reduce the AMIPS problem to an approximate nearest neighbor search (ANNS) problem [4, 29, 31, 47, 57–59], and further apply LSH indexes for addressing ANNS problem. In contrast, graph-based methods [20, 34, 56] builds an approximate proximity graph (APG) to facilitate AMIPS. In the APG, each vertex represents a data point in the dataset and each edge connects two vertices with a smaller distance. Based on APG, AMIPS can be solved by adopting beam search in the APG [20, 34, 56]. However, existing solutions suffer from the following three challenges.

(C1) *Inadequate query accuracy.* Graph based methods are well known for their high query accuracy in the approximate nearest neighbor search (ANNS) problem. This effectiveness stems from the triangle inequality in metric spaces, which ensures that (i) *the neighbor of a neighbor is very likely to be a neighbor*, and (ii) *the nearest neighbors of a query point are in close proximity and can be reached from one another in the approximate graph with only a few hops*. Consider the APG shown in Figure 1(a), which includes points x_1 , x_2 and x_3 in a metric space. Here, x_1 is a neighbor of x_2 , and x_2 is the neighbor of x_3 . Consequently, x_1 is also a neighbor of x_3 . For the query point q_2 , its two neighbors, x_1 and x_3 , are located within a small region of the metric space.

However, when applied to the AMIPS problem, performance deteriorates because the aforementioned properties do not hold in inner product spaces, where the inner product is non-metric. As a result, the common search strategy in APGs used for ANNS becomes inefficient for AMIPS. Figure 1(b) illustrates the APG in an

¹In this paper, we use the terms “points” and “vectors” interchangeably.

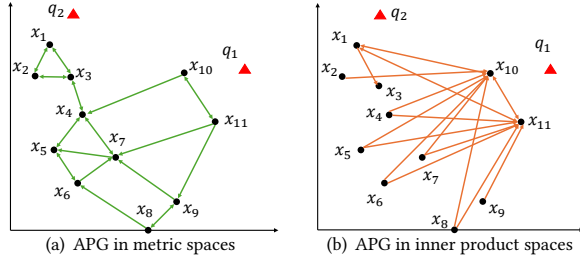


Figure 1: APG in metric spaces and inner product spaces.

inner product space. Observe that x_9 is a neighbor of x_{11} , and x_{11} is a neighbor of x_1 , but x_9 is not a MIPS neighbor of x_1 due to their large angular distance. Additionally, the AMIPS problem suffers from *norm bias* issues, *i.e.*, points with large norms are more likely to be MIPS neighbors of other points. Consequently, points with large norms tend to be mutually connected, making it difficult for a point with modest norms to be reached even if they are promising MIPS neighbors of the query point. For example, in Figure 1(b), x_1 , x_{10} and x_{11} are mutually connected and do not regard other points as a MIPS neighbor. Once the search enters into x_1 , x_{10} and x_{11} , we cannot find other points. Compared to graph based methods, LSH based methods achieve a worse query accuracy due to the *hash-boundary issue* [21, 48, 56]: some points that are close in the original space may be mapped into different hash buckets, thus not being identified by LSH-based approaches as neighbors [56].

(C2) *Low query efficiency.* For graph-based methods, unlike in metric spaces, the MIPS results of a query point q are often widely distributed in the inner product space, while two inner product neighbors of q may have a small inner product with each other. This leads to the APG search encountering multiple disjoint local maxima, making efficient convergence difficult. As shown in Figure 1(b), although q_1 identifies x_1 and x_9 as two of its MIPS neighbors, these points are positioned on opposite sides of q_1 and have an extremely small inner product with each other. LSH-based methods address the AMIPS problem by accessing the hash buckets near q across multiple hash tables. However, due to their poor estimation quality, they require significantly more time to achieve the same level of accuracy as graph-based methods.

(C3) *Limited scalability for large-scale dataset.* The limited scalability of LSH-based and graph-based methods for large-scale datasets arises from different challenges. For LSH-based methods, the number of points in a hash bucket increases with data cardinality, leading to a larger candidate size during query processing. In the worst cases, the query cost of some LSH-based methods [25, 57] can approach $O(n)$, where n is the cardinality of the datasets, making them impractical for large-scale datasets. On the other hand, graph-based methods face significant construction costs, as they require identifying appropriate neighbors for all points to build the APG. Additionally, the time needed to find the neighbors of a point also grows with data cardinality.

In this paper, we present a novel graph-based framework called MARIA (approximate maximum inner product search in vector datasets) for efficiently answering the AMIPS problem on large-scale vector datasets. Specifically, MARIA first partitions a dataset into different blocks based on norms of points and constructs an APG for each block to capture inner product relationships, thereby

alleviating norm bias issues. Searching within a block using the APG is much more efficient and accurate than searching the entire dataset, as the points within a block ensure that a MIPS neighbor of the query's MIPS neighbor is likely to be the query's MIPS neighbor when the points in a block have similar norms. Then, MARIA connects all APGs by linking points across different APGs based on their angular similarity, facilitating fast AMIPS queries on inter-block APGs.

In addition, we develop three optimization techniques to further enhance the query performance and scalability of MARIA. First, we design a lightweight yet efficient LSH index to mitigate the significant cost of constructing the novel APG from scratch. For each point, we retrieve angular neighbors from the LSH index, which serve as entry points for refining the APG. These entry points streamline the query process for identifying high-quality neighbors for each point, thereby reducing construction costs and improving support for large-scale datasets. Second, we implement a skip-list-like strategy to selectively choose inter-block edges, thereby reducing the overall size of the APG. This approach significantly decreases the construction cost and index size of inter-block edges while still ensuring query accuracy, improving support for large-scale datasets. Finally, to further boost query efficiency, we propose a compact yet accurate LSH-based quantization technique to quickly estimate the inner product of two points, allowing us to filter out unqualified points during the search. This significantly reduces the number of points accessed during the graph search.

The major contributions of this paper are as follows.

- We provide a comprehensive analysis of state-of-the-art LSH-based and graph-based methods for AMIPS, and identify the challenges that highlight the inefficiencies of existing approaches.
- We propose a novel graph-based framework called MARIA with an innovative graph structure to efficiently address the AMIPS problem. MARIA not only alleviates the norm bias issue in the MIPS problem but also captures angular similarity to facilitate fast AMIPS queries.
- We present three optimizations to further enhance the query efficiency and scalability of our MARIA framework. In particular, we design a lightweight yet efficient LSH index to mitigate the cost of constructing an APG and develop a skip-list-like strategy to reduce the overall size of the APG, along with an LSH-based quantization technique to lower query costs. Additionally, we provide a theoretical analysis of query quality and computational complexity.
- Extensive experiments on real-world datasets demonstrate that MARIA outperforms existing methods in terms of query efficiency, accuracy, and scalability for large-scale datasets.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 presents the preliminaries. Section 4 introduces the MARIA framework, while Section 5 discusses the optimization techniques. Section 6 provides a theoretical analysis. Section 7 presents experimental studies that offer insights into the performance of the proposed MARIA and its main competitors. Finally, we conclude the paper in Section 8.

2 Related Work

The (approximate) Maximum Inner Product Search (MIPS) problem is a well-established problem in the database community. Solutions to the AMIPS problem predominantly utilize methods based on locality-sensitive hashing (LSH) [5, 25, 42, 43, 53, 57] and graph-based methods [11, 32, 35, 45, 46, 60].

2.1 LSH based Methods

An LSH function, denoted as $h(\cdot)$, is characterized by the property that for any two points p and q , the probability of collision, $\Pr[h(p) = h(q)]$, increases with the similarity between p and q [3, 21, 22]. By projecting high-dimensional points into lower-dimensional spaces using a set of LSH functions, it becomes feasible to identify high-quality results with a high probability by examining only the points near the query point in the projected spaces [3, 22]. However, since the inner product is not a metric, it has been demonstrated that no hashing function is inherently locality-sensitive for inner products [42, 43]. This indicates that for any query point q , there always exists another x such that $q^\top x > q^\top q$. Under any hash function $h(\cdot)$, the probability $\Pr[h(x) = h(q)]$ will not exceed $\Pr[h(q) = h(q)] = 1$, contradicting the principle that collision probability should increase with similarity. To overcome this issue, methods like L2-ALSH [43] and Sign-ALSH [42] employ asymmetric transformations to convert the AMIPS problem into an ANNS problem, subsequently applying LSH-based methods to solve it [5, 42, 43]. This approach has led to the development of several LSH-based methods with asymmetric transformations, such as Simple-LSH [36], Range-LSH [53], H2-ALSH [25], and FARGO [57], which are favored for their straightforward structure and superior performance. To exploit the norm bias that the point with large norms are more likely to be a MIPS neighbor of the query point, some methods partition the dataset into several blocks based on their norms and then apply asymmetric transformations within each block [25, 53, 57]. Although this strategy significantly enhances query accuracy, LSH-based methods still struggle to achieve high query efficiency due to the large candidate size resulting from multiple blocks.

2.2 Graph based Methods

Due to the superior query performance of graph-based methods in ANNS, many studies have adopted these methods for addressing the AMIPS problem [32, 35, 45, 46, 60]. HNSW [34], one of the most effective graph-based methods, approximates the Delaunay Graph [27] to efficiently solve ANNS. The Delaunay graph [27] is known to solve the NNS problem exactly. Therefore, existing graph-based methods aim to better approximate the Delaunay graph in the inner product space. Ip-NSW [34] addresses the AMIPS problem by replacing the similarity function in HNSW from *L2-distance* to *inner product*, demonstrating that a Delaunay graph in an inner product space can also ensure exact MIPS results. However, it remains unclear how closely the constructed ip-NSW resembles a true Delaunay graph, as ip-NSW is built heuristically. Ip-NSW⁺ highlights the limitations of ip-NSW, stemming from the differences between inner product and L2-distance, noting that a MIPS neighbor of a MIPS neighbor is unlikely to be a MIPS neighbor again (i.e., *challenge 2*). Consequently, the search in ip-NSW involves numerous unnecessary computations, as many edges in ip-NSW are unsuitable

for AMIPS. Ip-NSW⁺ enhances ip-NSW's performance by using an additional APG in angular space to generate entry points for continuous AMIPS searches. However, optimizing only the entry points offers limited improvements due to the large number of unsuitable edges in ip-NSW. Other works, such as Möbius-Graph [60], IPDG [46], and NAPG [45], focus on designing better strategies to approximate the Delaunay graph in an inner product space. Möbius-Graph employs the Möbius transformation to convert the inner product space into a Euclidean space, subsequently building HNSW in the transformed space. IPDG [46] and NAPG [45] demonstrate that Delaunay graphs in inner product and Euclidean spaces have different structures, leading them to design new edge selection strategies to approximate the Delaunay graph in the inner product space better. However, as an exact Delaunay graph in hundreds of dimensional spaces will be a very dense graph and is nearly impossible to be computed. These methods achieve limited performance since their structures are greatly different from an exact Delaunay graph.

2.3 Additional Methods

Additionally, other methods, such as learning, sampling, and quantization based methods, have been proposed for AMIPS [14, 19, 41]. Shen et al. [41] propose a binary code learning framework that preserves the inner product among raw data vectors. BanditMIPS [49] estimates the inner product for each point by subsampling coordinates and adaptively evaluates more coordinates for more promising points. CeoMIPS [38] selects a few projections associated with the extreme values of the query signature. However, these methods focus on reducing the cost of computing an inner product but have high query complexity to achieve high query accuracy, limiting their scalability to large-scale datasets.

3 Preliminaries

Let $\mathbb{R}^d$ be a d -dimensional vector space, $x_1^\top x_2$ the inner product between points $x_1, x_2 \in \mathbb{R}^d$, and $\langle x_1, x_2 \rangle$ the angle between x_1 and x_2 . Frequently used notations are summarized in Table 1.

3.1 Problem Definition

DEFINITION 1 (MAXIMUM INNER PRODUCT SEARCH). Given a dataset $\mathcal{D} \subseteq \mathbb{R}^d$ with $|\mathcal{D}| = n$ and a query point $q \in \mathbb{R}^d$, the maximum inner product search (MIPS) returns a point $x^* \in \mathcal{D}$:

$$x^* = \arg \max_{x \in \mathcal{D}} q^\top x. \quad (1)$$

DEFINITION 2 ((c, k)-APPROXIMATE MAXIMUM INNER PRODUCT SEARCH [55]). Given a dataset $\mathcal{D}$, a query point $q \in \mathbb{R}^d$, an approximation ratio $0 < c < 1$, and a positive integer k , a (c, k)-approximate maximum inner product search (AMIPS) returns k points $x_1, \dots, x_k \in \mathcal{D}$ sorted in descending order w.r.t. their inner products with q . For $\forall x_i, i = 1, \dots, k$, we have $q^\top x_i \geq c \cdot q^\top x_i^*$, where $q^\top x_i^*$ is the i -th maximum inner product.

When all points in $\mathcal{D}$ have the identical norm, the AMIPS problem is equivalent to the approximate nearest neighbor search (ANNS) problem under the angular distance.

DEFINITION 3 ((c', k)-APPROXIMATE NEAREST NEIGHBOR SEARCH). Given a dataset $\mathcal{D}$, a query point $q \in \mathbb{R}^d$, an approximation ratio $c' > 1$, and a positive integer k , a (c', k)-approximate nearest neighbor

Table 1: List of Key Notations

Notation	Definition
$\mathcal{D} \subseteq \mathbb{R}^d$	A dataset
$n = \mathcal{D} $	The dataset cardinality
x, u, v	A point (a.k.a. vector) in $\mathcal{D}$
$N(x)$	x 's neighbor set
q	A query point
$q^\top x$	The inner product between q and x
$\langle q, x \rangle$	The angle between q and x
s	The number of blocks
M	The maximum inner-block degree
M'	The maximum inter-block degree between two blocks

search (ANNS) returns k points $x_1, \dots, x_k \in \mathcal{D}$ sorted in ascending order w.r.t. their angular distance to q . For $\forall x_i, i = 1, \dots, k$, we have $\langle q, x_i \rangle \leq c' \cdot \langle q, x_i^* \rangle$, where x_i^* is the i -th nearest neighbor of q in $\mathcal{D}$.

Based on the definition above, it is clear that

$$\arg \min_{x \in \mathcal{D}} \langle q, x \rangle = \arg \max_{x \in \mathcal{D}} \frac{q^\top x}{\|q\| \|x\|} = \arg \max_{x \in \mathcal{D}} q^\top x, \quad (2)$$

where the first equality holds because of $\cos \langle q, x \rangle = \frac{q^\top x}{\|q\| \|x\|}$, and the second equality follows from the identical norm in $\mathcal{D}$. Therefore, solving the MIPS problem is equivalent to finding a point x with the smallest angular distance to q .

REMARK 1. For clarity, we simplify (c, k) -AMIPS and (c', k) -ANNS problems to c -AMIPS and c' -ANNS when $k = 1$.

EXAMPLE 1. For the dataset $\mathcal{D}$ in Figure 1, $n = |\mathcal{D}| = 11$. For the query point q_2 , its exact MIPS result in $\mathcal{D}$ is x_{10} and exact nearest neighbor in angular spaces is x_1 . Given $c = 0.8$, we can return one of the points x_1, x_2 and x_3 as the result of 0.8-AMIPS. Similarly, we can return x_1 and x_5 as its (1.5, 2)-ANNS results in angular spaces.

3.2 Locality Sensitive Hashing based Methods

Locality-sensitive hashing (LSH) is one of the most popular tools for solving the ANNS problem due to its simple structure, sublinear query cost, and rigorous quality guarantee. The formal definition of LSH is given below.

DEFINITION 4 (LOCALITY-SENSITIVE HASHING). Given a distance function dist , a distance threshold r , an approximation ratio $c > 1$, and two probability values p_1, p_2 , where $p_1 > p_2$, a family $\mathcal{H} = \{h : \mathbb{R}^d \rightarrow U\}$ is called (r, cr, p_1, p_2) -locality sensitive, if for any $x_1, x_2 \in \mathbb{R}^d$, it satisfies the following two conditions:

- (1) If $\text{dist}(x_1, x_2) \leq r$, then $\Pr[h(x_1) = h(x_2)] \geq p_1$;
- (2) If $\text{dist}(x_1, x_2) \geq cr$, then $\Pr[h(x_1) = h(x_2)] \leq p_2$.

A typical LSH for angular distance is defined as follows [8]:

$$h(x) = \text{sign}(a^\top x), \quad (3)$$

where a is a d -dimensional random vector with each entry independently chosen from a 2-stable distribution, i.e., the standard normal distribution $N(0, 1)$.

3.3 Graph based Methods

The graph-based methods build the approximate proximity graphs (APG) for vector similarity search.

DEFINITION 5. Given a similarity function $f : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and a dataset $\mathcal{D} \in \mathbb{R}^d$, an approximate proximity graph (APG) $\mathcal{G} = (V, E)$ is a graph where the vertex sets $V = \mathcal{D}$ and the edge sets $E \subseteq \{(x_1, x_2) | x_1, x_2 \in \mathcal{D}\}$. The edge $(x_1, x_2) \in E$ implies that x_2 has a high similarity to x_1 in $\mathcal{D}$ according to the function f .

REMARK 2. In a metric space $\mathcal{M} = (\mathbb{R}^d, \text{dist})$, we define the similarity function f as $f(x_1, x_2) = -\text{dist}(x_1, x_2)$.

Terminology. Given an APG $\mathcal{G} = (V, E)$, for an edge $(x_1, x_2) \in E$, we denote x_2 as the *neighbor* of x_1 . (x_1, x_2) is an *out-edge* of x_1 and an *in-edge* of x_2 . The set $N(x) = \{v | (x, v) \in E\}$ is the *neighbor set* of x in $\mathcal{G}$, while $RN(x) = \{u | (u, x) \in E\}$ denotes the *reverse neighbor set* of x in $\mathcal{G}$. We define $|N(x)|$ as the *out-degree* of x and $|RN(x)|$ as the *in-degree* of x . For clarity, we often refer to the out-degree of x simply as the *degree* of x .

REMARK 3. Graph-based methods typically do not explicitly utilize the approximation ratio c to control the quality of the query. To avoid confusion, we abbreviate (c, k) -AMIPS as k -AMIPS for simplicity.

3.4 Limitations of Existing AMIPS methods

Limitations of LSH based methods. First, LSH based methods achieve a worse query accuracy due to the *hash-boundary issue* [21, 48, 56]. Specifically, even when some points are close in the space, they can still be mapped to different hash buckets. As shown in Figure 3, x_3 and x_4 are mapped into different buckets in $T_{2,2}$ even though their angular distance is smaller than 30° . Second, LSH-based methods address the AMIPS problem by accessing the hash buckets near the query point across multiple hash tables. However, due to their poor estimation quality, they require significantly more time to achieve the same level of accuracy as graph-based methods. Third, the number of points in a hash bucket increases with data cardinality, leading to a larger candidate size during query processing. In worst cases, the query cost of some LSH based methods [25, 57] approaches $O(n)$, where n is the cardinality of the datasets. This makes LSH based methods impractical for large-scale datasets.

Limitations of graph based methods. Graph based methods are well known for their high query accuracy in ANNS problems. This effectiveness stems from the triangle inequality in metric spaces, which ensures that (i) *the neighbor of a neighbor is very likely to be a neighbor*, and (ii) *the nearest neighbors of a query point are in close proximity and can be reached from one another in the approximate graph with only a few hops*. However, when applied to the AMIPS problem, performance deteriorates because these properties do not hold in inner product spaces, where the inner product is non-metric. As a result, the common search strategy in APGs used for ANNS becomes inefficient for AMIPS. In addition, unlike in metric spaces, the MIPS results of a query point q are often widely distributed in the inner product space, while two inner product neighbors of q may have a small inner product with each other. This leads to the APG search encountering multiple disjoint local maxima, making efficient convergence difficult. Furthermore, graph-based methods face significant construction costs, as they require identifying appropriate neighbors for all points to build the APG.

4 The MARIA Framework

In this section, we introduce a novel graph-based framework called MARIA, designed to efficiently solve the AMIPS problem. An overview of MARIA is depicted in Figure 2. MARIA addresses the aforementioned challenges (*i.e.*, inadequate query accuracy, low query efficiency, and limited scalability) as follows.

Given a dataset $\mathcal{D}$, MARIA first adopts a *norm-based Partition* strategy (see Section 4.1) to partition $\mathcal{D}$ into blocks $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_s$ based on norms of points. For each block, MARIA constructs an APG (*i.e.*, inner-block APG) (see Section 4.2) to connect inner-block points to take advantage of the state-of-the-art performance of graph-based methods for ANNS. To support queries over all blocks, MARIA builds inter-block edges between two blocks $\mathcal{D}_i$ and $\mathcal{D}_j$ to linking points $u \in \mathcal{D}_i$ and $v \in \mathcal{D}_j$ based on the angular similarity of u and v . Based on the novel APG, MARIA can efficiently process the AMIPS query (see Section 4.3). Moreover, MARIA integrates three optimizations to further enhance query efficiency and scalability (see Section 5). Compared to existing solutions, MARIA has the following advantages:

- (i) *High-quality query results.* MARIA constructs an APG within each block, where points have similar norms, effectively mitigating norm bias issues. Searching within a block using the APG is more accurate than searching the entire dataset.
- (ii) *Fast query processing.* MARIA enables efficient searches by enhancing neighbor relationships within each APG. For multiple searches across all blocks, entry points generated in the LSH framework, along with query results from one block, guide searches in subsequent blocks, facilitating faster convergence. Additionally, a quantization-based pruning condition (see Section 5.1) is employed to further reduce computational costs.
- (iii) *Low construction cost.* By generating neighbor sets for each point through searches within its block, MARIA ensures low construction costs using efficient query strategies. Additionally, a skip-list-like mechanism (see Section 5.2) reduces the number of searches required during construction, further minimizing costs.
- (iv) *Desirable scalability.* Leveraging its novel graph structure and effective optimizations, MARIA achieves superior indexing efficiency and query performance, making it well-suited for large-scale datasets.

EXAMPLE 2. Consider $\mathcal{D}$ in Figure 2. With $\alpha = 0.7$, $\mathcal{D}$ is divided into three blocks: $\mathcal{D}_1 = \{x_1, x_{10}, x_{11}\}$, $\mathcal{D}_2 = \{x_2, x_3, x_4, x_7, x_8, x_9\}$, and $\mathcal{D}_3 = \{x_5, x_6\}$ via norm-based partition. An APG is constructed within each block, as shown by the green lines. Unlike the APG in Figure 1(b), where all edges are captured by x_1, x_{10} , and x_{11} , MARIA allows each point to be probed, mitigating norm bias issues. Inter-block edges, depicted by the blue lines (e.g., (x_1, x_2)), connect different blocks. To perform a 2AMIPS query for q_2 , MARIA starts with block $\mathcal{D}_1$ and finds x_1 . It then proceeds to search in $\mathcal{D}_2$ starting from x_2 , as x_2 is the inter-block neighbor of x_1 , significantly reducing the search cost in $\mathcal{D}_2$. After searching these blocks, x_1 and x_2 are returned as the 2AMIPS results.

4.1 Norm-based Partition

AMIPS results are influenced by both vector norms and their angles relative to the query, leading to a wide distribution of results that makes constructing a single APG challenging. To handle this,

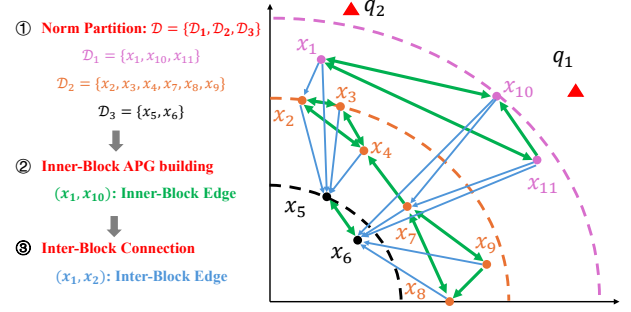


Figure 2: The MARIA framework

MARIA introduces a norm-based partitioning strategy, which divides the dataset into multiple blocks based on the norms of vectors. Vectors within each block have similar norms. This approach allows MARIA to treat the AMIPS problem within each block as equivalent to the ANNS problem in the angular space. As shown in Figure 2, all data points are partitioned into three blocks $\mathcal{D}_1, \mathcal{D}_2$ and $\mathcal{D}_3$ illustrated with different colors. Let $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_s$ denote partitioned blocks, where the points in $\mathcal{D}_1$ have the largest norms. We adopt a simple but effective partition strategy that focuses on controlling the norm range in a block. Specifically, given a ratio $\alpha \in (0, 1)$, for each block $\mathcal{D}_i$, let x_1, x_2 be any two points in $\mathcal{D}_i$, it satisfies $\alpha \cdot \|x_1\| < \|x_2\|$. This method effectively divides the space into onion-like concentric hyperspheres, with each block representing a hyperring of vectors.

LEMMA 1. Given a norm ratio α , a correct angular neighbor of the query point q in a block $\mathcal{D}_i$ is the α -AMIPS neighbor of q .

The proof of this lemma is provided in Section 6.1. We have demonstrated the AMIPS problem becomes equivalent to the ANNS problem when all points have the identical norm. Lemma 1 reveals that the equivalence still exists in a block. Therefore, the AMIPS problem can be solved more accurately via the APG in a block than in the whole dataset since the block satisfies the assumption that the neighbor of a neighbor is very likely to be a neighbor.

4.2 APG construction in MARIA

After partitioning $\mathcal{D}$ into blocks $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_s$, MARIA constructs an APG for each block and then connects any two of blocks $\mathcal{D}_i$ and $\mathcal{D}_j$ by linking points $u \in \mathcal{D}_i$ and $v \in \mathcal{D}_j$ based on the angular similarity of u and v .

4.2.1 Inner-Block Construction. The APG in the block $\mathcal{D}_i$, denoted as $\mathcal{G}_i = (V_i, E_i)$, is a directed MIPS graph. Here, $E_i = \{(u, v) | u, v \in \mathcal{D}_i\}$ is the edge set where v is an AMIPS neighbor of u in $\mathcal{D}_i$. Building the APG $\mathcal{G}_i$ is extremely time-consuming since we are required to find MIPS neighbors for each point. There are mainly two commonly used strategies that are designed to lower the construction cost. In the first strategy: NN-Descent [18, 23, 37], the APG is improved from an initial APG iteratively. In the initial APG, each vertex is attached with M neighbors randomly. At one epoch, for any two neighbors u and v of a point x , we compute $u^\top v$ and update the neighbor sets of u and v , refining the neighbor quality of u and v . The second strategy builds an APG via consecutive insertion strategy where the point is inserted into the APG one by one, as that in ip-NSW [35] and LSH-APG [56]. When inserting a point x ,

we conduct a MIPS search for x to find its neighbors in the current APG to generate its edge sets. The cost of the consecutive insertion strategy depends on the efficiency of the MIPS search in the APG. Although NN-Descent is demonstrated to build a higher-quality APG [12], the built APG via NN-Descent achieves a limited query performance due to the low connectivity [20] and unified degree M [56]. Low connectivity makes many points unreachable from other vertex and the unified degree fails to consider the varied distribution in the real-world dataset. Therefore, we choose the consecutive insertion strategy to build a varied-degree APG in $\mathcal{D}_i$ for a better query performance.

The construction is described in Algorithm 1. We build $\mathcal{I}_G$ via the consecutive insertion strategy as described in Algorithm 1. To insert a point x into $\mathcal{G}_i$, we find its M_0 AMIPS results in $\mathcal{D}_i$ as its neighbor set $N(x)$ (Line 4). To find the M_0 AMIPS results, we conduct a beam search for x in the current APG of $\mathcal{D}_i$, as described in Algorithm 3. The details of search is explained in Section 4.3. After finding $N(x)$, we insert these edges into $\mathcal{G}_i$ (Line 6) and update the neighbor set for each point v in $N(x)$ with x . Each point v stores at most M AMIPS results in its neighbor set (Line 8-10). In the final APG, the degree of each point varies in a range $[M_0, M]$. All the points are allocated with M_0 neighbors once inserted. Then, once it is selected as the AMIPS neighbor of another points, it can append a neighbor. Therefore, the points that are liable to be the MIPS result of other points are likely to have more neighbors. Such a varied-degree strategy fully considers the difference among the data distribution and facilitates the efficient search. The other parameter efC is adopted for controlling the query quality where a larger efC achieves a higher quality but incurs a larger query cost. And M controls the maximum degree because a too large degree will increase the query cost in turn [50, 56].

Algorithm 1: Building APG in a block

Input: A block $\mathcal{D}_i$, parameters efC , M_0 and M
Output: An APG in $\mathcal{D}_i$

```

1  $E_i \leftarrow \emptyset$ ;
2  $N_I \leftarrow$  random  $M_0$  inserted points in  $\mathcal{D}_i$ ;
3 for each point  $x \in \mathcal{D}_i$  do
4    $N_f \leftarrow$  Call AMIPS( $x, \mathcal{G}_i, N_I, N_I, efC$ );
5   for each  $v \in N_f$  do
6      $E_i \leftarrow E_i \cup \{(x, v)\}$ ;
7      $E_i \leftarrow E_i \cup \{(v, x)\}$ ;
8     if  $|N(v)| > M$  then
9        $u^* \leftarrow \arg \min_{u \in N(v)} u^\top v$ ;
10       $E_i \leftarrow E_i - \{(v, u^*)\}$ ;
11  $\mathcal{G}_i \leftarrow \{\mathcal{D}_i, E_i\}$ ;
12 return  $\mathcal{G}_i$ ;
```

4.2.2 Inter-Block Connection. For clarity, we define two types of edges in the graph. The first type, *inner-block edges*, represents the edges built within each block, as described in Section 4.2.1. These edges capture the inner product relationships between points in the same block, alleviating norm bias issues due to their similar norms. The second type, *inter-block edges*, connects data points across blocks. Let $\mathcal{D}_1$ represent the partition with the largest norms, we

Algorithm 2: Inter-Block Connection

Input: APGs $\{\mathcal{G}_i = (\mathcal{D}_i, E_i)\}_{i=1:s}$ and parameters M', efC
Output: An APG $\mathcal{I}_G$

```

1  $\mathcal{I}_G \leftarrow \cup_{i=1}^s \mathcal{G}_i$ ;
2 for  $i \leftarrow 1$  to  $s$  do
3   for  $j \leftarrow i + 1$  to  $s$  do
4     for each  $u \in V_i$  do
5        $N_I \leftarrow$  random  $M'$  inserted points in  $\mathcal{D}_j$ ;
6        $N_f \leftarrow$  Call AMIPS( $u, \mathcal{G}_j, N_I, M', M'$ );
7       for each  $v \in N_f$  do
8          $\mathcal{I}_G \leftarrow \mathcal{I}_G \cup \{(u, v)\}$ ;
9 return  $\mathcal{I}_G$ ;
```

Algorithm 3: AMIPS($q, \mathcal{G}, EPs, R, k, ef$)

Input: A query point q , the APG $\mathcal{G}$, the initial entry point EPs , the result sets R , the parameters k and ef
Output: k points in $\mathcal{G}_i$

```

1 while  $|EPs| > 0$  do
2    $e_p \leftarrow \arg \max_{u \in EP} q^\top u$ ;
3   Remove  $e_p$  from  $EP$ ;
4    $x_f \leftarrow \arg \min_{u \in R} q^\top u$ ;
5   if  $|R| \geq ef$  and  $q^\top e_p < q^\top x_f$  then
6     break;
7   for  $x \in N(e_p)$  do
8     if  $x$  has not been visited then
9       Update  $x$  into  $EPs$  and  $R$ ;
10      while  $|R| > ef$  do
11         $x' \leftarrow \arg \min_{u \in R} q^\top u$ ;
12        Remove  $x'$  in  $R$ ;
13  $R \leftarrow$  The best  $k$  AMIPS results to  $q$  in  $R$ ;
14 return  $R$ ;
```

demonstrate the procedure of finding inter-block edges in Algorithm 2. Specifically, for each two blocks $\mathcal{D}_i$ and $\mathcal{D}_j$ where $i < j$ and any a point $u \in \mathcal{D}_i$, we find M' angular neighbors for u in $\mathcal{D}_j$. The inter-block edges are unidirectional and they are always incident from the point with a larger norm to the point with a smaller norm. Since the search always begins from $\mathcal{D}_1$ and an inter-block edge (u, v) servers as the guidance for quickly determining the high-quality candidate v for the search in v 's block once we find the MIPS neighbor u in u 's block.

4.3 AMIPS Query Processing

The AMIPS search in MARIA conducts a series of search block by block. As shown in Algorithm 3, the AMIPS query in a block starts from an initial entry point set EPs and generate the AMIPS result for the query point q via a cyclic probing in the APG. In each epoch, the point e_p that has the maximum inner product to q in EPs is selected as the entry point (Line 2). For each neighbor x of e_p in the APG, we compute its inner product to q and update it into sets EP and R . The result set R stores the best ef points found so far. The search procedure repeats until we cannot find a point in EPs that has a larger inner product to q than any of the best ef points in R .

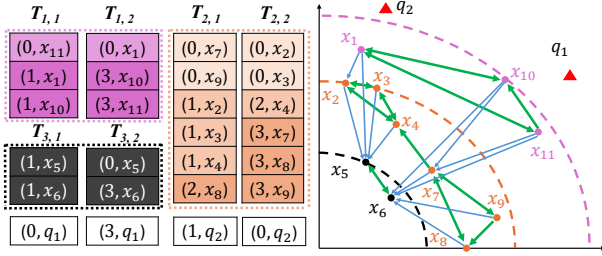


Figure 3: LSH-based Initialization

Algorithm 4: k ANN Query($q, \{\mathcal{G}_i\}, k, ef$)**Input:** A query q , the APGs $\{\mathcal{G}_i\}$ and parameters k and ef **Output:** k points in $\mathcal{I}_G$

```

1  $N_I \leftarrow$  random  $M_0$  points in  $\mathcal{D}_1$ ;
2  $R \leftarrow N_I$ ;
3 for  $i \leftarrow 1$  to  $s$  do
4   if  $|R| \geq ef$  and  $\|q\| \cdot \max_{x \in \mathcal{D}_i} \|x\| < \min_{x \in R} q^\top x$  then
5     break;
6   end
7    $R \leftarrow$  Call AMIPS( $q, \mathcal{G}_i, N_I, R, ef$ );
8    $N_I \leftarrow \{v \in \mathcal{D}_{i+1} | u \in R, (u, v) \in \mathcal{I}_G\}$ ;
9 end
10 return  $R$ ;
```

(Line 5). Then, the search terminates and we return k best points in R as the AMIPS results of q . The parameter ef is set to balance the query efficiency and accuracy. A higher ef makes the termination harder (Line 5), which allows us to access more points to refine the query results but incurs a higher query cost.

To enhance the query efficiency, the search among different blocks are not independent. As described in Algorithm 4, the search begins from the block $\mathcal{D}_1$ since the points here have the largest norms. Then, we adopt the more accurate initial set EPs and result set R for the search in the next blocks to guide the search better (Line 7). For the search in $\mathcal{D}_{i+1}$, the initial result set R consists of the best k points found in the former i blocks, which enables the search terminate faster if there are few qualified results in $\mathcal{D}_{i+1}$. Moreover, we refine the initial entry point set EPs by accessing the inter-block neighbors of the points in R . For a point u in R , u 's neighbors in $\mathcal{D}_{i+1}$ will be chosen as the entry points. Generating higher-quality entry points greatly reduces the hop numbers required to terminate the algorithm, and thus improve the query efficiency. Moreover, a closer entry point decreases the probability that the query terminates at a local maximal x where $q^\top x$ is still very small.

5 Optimizations

In this section, we introduce several optimization techniques to further enhance MARIA's performance, including (a) an LSH-based initialization; (b) a skip-list-like mechanism; and (c) a quantization and pruning technique.

- **LSH-based Initialization:** In section 4.2, the APG for each block was constructed using random initialization for search in the APG, which could result in high cost to converge the search, enlarging the construction costs. To address this, we introduce a lightweight LSH-based framework to efficiently select initial

Algorithm 5: The MARIA framework

```

1  $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_s \leftarrow$  Partition  $\mathcal{D}$  to blocks;
2  $\mathcal{I}_H \leftarrow$  Call LSHInit;
3 for each  $\mathcal{D}_i$  do
4   Generate initial neighbor sets in  $\mathcal{I}_{H_i}$  for each point;
5    $\mathcal{I}_{G_i} \leftarrow$  Call BuildingAPG;
6  $\mathcal{I}_H \leftarrow$  the union of all  $\mathcal{I}_{H_i}$ ;
7  $\mathcal{I}_G \leftarrow$  Call Inter-Block Connection;
8 return  $\mathcal{I}_H$  and  $\mathcal{I}_G$ ;
```

neighbors for each vertex, reducing the cost of search in an APG for generating the neighbor sets.

- **Skip-List-like Mechanism:** In Section 4.2.2, we introduce a naive inter-block connection mechanism, which adds notable complexity and inefficiency. To optimize this, we propose a novel skip-list-like strategy that reduces the number of inter-block edges and redistributes more inter-block edges to the blocks that are close to each other, significantly mitigating the index size and construction cost without scarifying the query performance.
- **Quantization and Pruning:** To further boost search efficiency, MARIA incorporates a quantization technique that exploits the properties of LSH to estimate inner products between points. This allows for the efficient pruning of unqualified candidates during the search, reducing the number of distance computations needed and significantly accelerating the query processing.

The complete MARIA framework, incorporating all optimizations, is outlined in Algorithm 5.

5.1 LSH-based Initialization

5.1.1 Building LSH framework. $\mathcal{I}_H$ consists of L hash tables for each block $\mathcal{D}_i$. In each hash table T_l , we randomly select K hash functions, $h_{l,1}, \dots, h_{l,K}$ from the hash family $\mathcal{H}$, as defined in Equation 3. Then, for a point x , we compute its K hash values. As $h_{l,j}(x) \in \{0, 1\}$, we can transform the K -bit hash values of x , $h_{l,1}, \dots, h_{l,K}$, into its decimal representation $g_l(x)$, i.e.,

$$g_l(x) = \sum_{j=1}^K h_{l,j}(x) \cdot 2^{K-j}. \quad (4)$$

Then, we store the pair $(g_l(x), x)$ as a *key-value* pair into the hash table T_l . Each block $\mathcal{D}_i$ has such a hash table $\{(g_l(x), x)\}_{x \in \mathcal{D}_i}$ where the $(g_l(x), x)$ pairs are sorted in ascending order of the keys. This process is repeated L times to build L hash tables for the block $\mathcal{D}_i$, denoted as $\mathcal{T}_i = \{T_{i,1}, \dots, T_{i,L}\}$. Algorithm 6 describes the procedure of building our LSH framework.

5.1.2 Search in LSH tables. The hash family $\mathcal{H}$ is designed under angular space. It satisfies:

LEMMA 2. Given any two point x_1 and x_2 , let $\theta = \langle x_1, x_2 \rangle$, for a hash function $h \in \mathcal{H}$ defined in Equation 3, the collision probability of $h(x_1)$ and $h(x_2)$ is computed as follows [36, 57].

$$\Pr[h(x) = h(q)] = 1 - \theta/\pi \quad (5)$$

Lemma 2 indicates that the points with small angular distances are more likely to have the same hash value under h . Hence, Upon the LSH tables, we can perform an ANNS query in the angular space

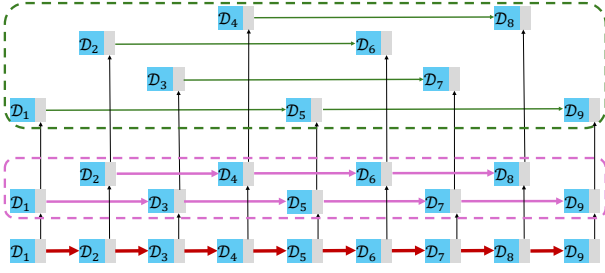


Figure 4: The Illustration of Skip-List-like Mechanism

for yielding a initial entry point set for each point $x \in \mathcal{D}_i$. To find the ANNS results in $\mathcal{D}_i$ of the point x , we select C candidates v that $g_l(x)$ is close to $g_l(v)$ from L table $T_{i,l}$. As each table is sorted, we can generate the candidates for all points by traversing in $T_{i,l}$ once (Line 9). Therefore, the cost of search in LSH tables is very small. After that, we verify all C candidates of x and choose the best M_0 results as its neighbors. These neighbors serve as the initial set for building the APG, replacing the random initialization in line 2 of Algorithm 1 and line 5 of Algorithm 2, while all other steps remain identical. Since the LSH-based initialization offers a higher quality and more stable starting point compared to random initialization, it significantly mitigates the high computational cost associated with graph construction, leading to faster convergence and improved overall graph quality.

EXAMPLE 3. Figure 3 illustrates the LSH tables in MARIA. Here, $L = 2$ and $K = 2$. Take x_7 as example, $g_1(x_7) = 0$ and $g_2(x_7) = 3$, so x_7 are stored as $(0, x_7)$ in T_1 and $(3, x_7)$ in T_2 . The points in different blocks are stored separately. To generate the initial set for points in $\mathcal{D}_2$, we can simply traverse T_1 and T_2 for $\mathcal{D}_2$ and select the proximity points in the hash table as the candidates to each other. After traversing T_1 , x_7 obtains the candidates x_9 . Similarly, x_7 obtains the candidates x_4 and x_8 in T_2 . By verifying x_4 , x_8 and x_9 , we insert x_4 and x_8 into the initial set.

Algorithm 6: LSHInit

Input: Blocks $\{\mathcal{D}_i\}$, Parameters L, K
Output: LSH index I_H

```

1 Generate  $L \cdot K$  hash functions  $\{h_{l,j}\}$ ;
2 for each  $\mathcal{D}_i$  do
3   for  $x \in \mathcal{D}_i$  do
4     Compute all  $L$  of  $g_l(x)$  values based on Eq. 4;
5     Insert  $(g_l(x), x)$  into hash table  $T_l$ ;
6   Sort all  $L$  tables  $\{T_l\}$ ;
7    $\mathcal{T}_i \leftarrow \{T_{i,1}, \dots, T_{i,L}\}$ ;
8  $I_H \leftarrow \{\mathcal{T}_1, \dots, \mathcal{T}_s\}$ ;
9 Traverse  $I_H$  to generate the initial set for each point;
10 return  $I_H$ 

```

5.2 Skip-List-like Mechanism

In Algorithm 2, each point in a block $\mathcal{D}_i$ is connected to $s - i$ other blocks. This would lead to excessive growth in the number of edges, significantly increasing the size of the APG and negatively impacting query efficiency. Moreover, the blocks with the smaller norms are less likely to have the expected AMIPS points than the

blocks with larger norms due to the norm bias. Therefore, the fully connected structure improves little over the query quality. To mitigate these issues, we draw inspiration from the skip list structure and propose a novel inter-block connection mechanism, which is designed as follows:

- (1) *Skip-list-like connection.* For a point x in $\mathcal{D}_i$, it connects to at most $\lfloor \log(s - 1) \rfloor$ blocks $\mathcal{D}_j$ where $j - i$ is the power of 2.
- (2) *Redistributed edges.* For a point x in $\mathcal{D}_i$, the number of its edges to the block $\mathcal{D}_j$ is $\lceil M'/(j - i) \rceil$.

LEMMA 3. The skip-list-like mechanism reduces the maximum inter-block degree of a point x from $O(M's)$ to $O(M' + \log s)$ and the cost of constructing the inter-block edges from $O(C_Q s)$ to $O(C_Q \log s)$ where C_Q is the cost of search in a block.

The proof of this lemma is provided in Section 6.1. This lemma demonstrate that the skip-list-like mechanism greatly reduces the index size and construction cost. Specifically, the *skip-list-like connection* ensures that the closer are two blocks, the higher is the probability that they are connected, as that in a skip list. Compared to the previous mechanism that the inter-block edges incident from a block $\mathcal{D}_i$ are uniformly distributed in the level of blocks, The *redistributed edges* strategy provides more inter-block edges to the points closer to $\mathcal{D}_i$. Intuitively, if a block $\mathcal{D}_j$ is closer to $\mathcal{D}_i$ than $\mathcal{D}_l$, it is earlier to be accessed than $\mathcal{D}_l$. So, it is more beneficial to generate the high-quality candidates by using a large number of inter-block edges. Moreover, in the previous strategy, the points in block $\mathcal{D}_j$ has $(j - 1) \cdot M'$ inter-block in-edges. Then, the blocks with smaller norm have more in-edges instead, which helps little for the query performance.

EXAMPLE 4. Figure 4 provides an example of our skip-list-like mechanism with $s = 9$. Here, $\mathcal{D}_1$ connects to only 3 blocks and the furthest block connected by it is $\mathcal{D}_5$. Given $M' = 4$, a point x in $\mathcal{D}_1$ has $4 + 2 + 1 = 7$ inter-block neighbors. More than 80% of these neighbors connect to the points in $\mathcal{D}_2$ and $\mathcal{D}_3$, which are more promising to be the AMIPS neighbor of a query. In the previous strategy, x has $(s - 1) \cdot M' = 32$ neighbors, 4 times larger than that in the skip-list-like mechanism. However, a half of them connect to $\mathcal{D}_6$, $\mathcal{D}_7$, $\mathcal{D}_8$ and $\mathcal{D}_9$, which are less likely to have the AMIPS results. Therefore, the skip-list-like mechanism greatly enhances the utilization ratio of inter-block edges.

5.3 Quantization and Pruning

The query cost of a graph-based method comes from the number of distance computations between q and the neighbors of e_p when accessing e_p . The previous query strategy described in Section 4.3 checks all e_p 's neighbors, which is unnecessary and time-consuming since some neighbors are obviously the unqualified candidates. To address this inefficiency, we introduce a pruning strategy that takes advantage of the properties of LSH functions.

5.3.1 Quantization. Consider that we have $m = L \cdot K$ hash functions, we concatenate all m hash values $h_{l,j}(x)$ for the point x as its quantization $Q(x)$. Then, we can estimate $\theta = \langle q, x \rangle$ for two points by using their quantization values.

LEMMA 4. For any two points q and x , denote $\theta = \langle q, x \rangle$ and

$$\beta(q, x) = \sum_{j=1}^m \mathbb{I}(Q_j(q) \neq Q_j(x)) \quad (6)$$

where $Q_j(x)$ is the j -th bit of $Q(x)$, then $\theta' = \frac{\pi\beta(q, x)}{m}$ is an unbiased estimator of θ .

Lemma 4 indicates that we can estimate θ accurately by using θ' when m is large enough as θ' is an unbiased estimator of θ . Since $q^\top x = \|q\| \cdot \|x\| \cdot \cos \theta$, we can further estimate $q^\top x$ before computing it. $\beta(q, x)$ can be quickly computed by the bitwise XOR operation. So, we can estimate $q^\top x$ efficiently and accurately.

5.3.2 Pruning Condition. Given a tolerant error $0 < \delta < 1$, the pruning condition based on the quantization values is as follows:

$$I_{\min} > I_{ub} = \|q\| \cdot \|x\| \cdot \cos(\theta' - z\sqrt{\theta'(\pi - \theta')/m}), \quad (7)$$

where $I_{\min}$ is the minimal inner product between q and the points in the result set, z is the quantile of the standard normal distribution at $1 - \delta/2$ to control the estimation error and efficiency. The intuition of the pruning condition is that when $I_{\min}$ is greater than I_{ub} , $I_{\min}$ will also be greater than $q^\top x$ with high probability $1 - \delta$. Then, it is reasonable to discard x without computing $q^\top x$. The correctness of this pruning condition is proven in Section 6. With a bit of estimation cost, this pruning condition significantly reduces the computation cost and still guarantees the query quality.

EXAMPLE 5. As shown in Figure 3, $m = L \cdot K = 4$. Take x_4 as example, $Q(x_8) = [g_1(x_8); g_2(x_8)] = [2; 3] = 1011$ and $Q(q_1) = [0; 3] = 0011$. So, $\beta(q_1, x_8) = 1$ and we can estimate $\theta = \langle q_2, x_4 \rangle$ as $\theta' = \pi\beta(q, x)/m = \pi/4$. Assume that $I_{\min} = q_1^\top x_{11} = 139$, $x_8 = (6, 0)$ and $q_1 = (11, 8)$, let z be $1/\sqrt{3}$, then I_{ub} can be estimated as 75.4. Hence, we can prune x_8 without computing $q^\top x_8$.

Algorithm 7: MARIA Search

Input: A query q , the APG I_G and parameters k and ef

Output: k points in I_G

/* Replace line 1 in Algorithm 4 */

1 $EP \leftarrow M_0$ points identified by I_H on $\mathcal{D}_1$;

/* Add after line 10 in Algorithm 3 */

2 **if** $I_{\min} > I_{ub}$ **then**

 /* Same as Line 8 - 12 in Algorithm 3 */

6 Theoretical Analysis

We first provide the proofs of lemmas in Sections 4 and 5. Then, we analyze the space and time complexity of our algorithm.

6.1 Proofs

We prove Lemma 1 that reveals the equivalence between the AMIPS and ANNS in a block.

PROOF TO LEMMA 1. Denote x^* as the exact angular neighbor of q in the block $\mathcal{D}_i$ and y^* as the exact MIPS neighbor of q in the block $\mathcal{D}_i$, $\theta_1 = \langle q, x^* \rangle$ and $\theta_2 = \langle q, y^* \rangle$, we have $\theta_1 < \theta_2$ and $\cos \theta_1 \geq \cos \theta_2$. Given the definition of the norm partition, we have $\|y^*\| \leq x^*/\alpha$. Hence, $q^\top y^* = \|q\| \cdot \|y^*\| \cdot \cos \theta_2 \leq \|q\| \cdot x^*/\alpha \cdot \cos \theta_1 = q^\top x^*/\alpha$. So, x^* is a correct α -AMIPS neighbor of q . $\square$

Then, we prove the Lemma 3, which demonstrates the advantages of the skip-list-like mechanism in terms of both index size and construction cost.

PROOF TO LEMMA 3. In the naive inter-connection strategy, each point in a block $\mathcal{D}_i$ will connect to other $(s - i)$ blocks and requires $(s - i)$ searches. So, the average inter-block degree is $s/2 \cdot M' = O(sM')$ and the average search cost is $O(C_Q s)$. However, in the skip-list-like mechanism, each point in a block $\mathcal{D}_i$ will connect to other $O(\log s)$ blocks, so the average search cost is $O(C_Q \log s)$. The maximum inter-block degree is

$$\sum_{l=1}^{\lceil \log M' \rceil} \frac{M}{2^{l-1}} + \lceil \log(s - 1) \rceil - \lceil \log M' \rceil < 2M' + \lceil \log s \rceil,$$

which is $O(M' + \log s)$. $\square$

Next, we analyze the property of $\beta(q, x)$ and prove Lemma 4 that reveals the effectiveness of the quantization.

LEMMA 5. Given m hash function $h_j \in \mathcal{H}$, for two points q and x , denote $\theta = \langle q, x \rangle$, $\beta(q, x) = \sum_{j=1}^m \mathbb{I}(h_j(q) \neq h_j(x))$ follows the binomial distribution $B(m, \theta/\pi)$ [16].

PROOF. According to Lemma 2, $\Pr[h_j(q) \neq h_j(x)] = \theta/\pi$. As all of m hash functions are independent, $\beta(q, x)$ follows $B(m, \theta/\pi)$ based on the definition of the binomial distribution [16, 17, 52]. $\square$

PROOF TO LEMMA 4. As $\beta(q, x) \sim B(m, \theta/\pi)$, $\mathbb{E}[\beta(q, x)] = m \cdot \theta/\pi$. Therefore, $\mathbb{E}[\theta'] = \frac{\pi}{m} \mathbb{E}[\beta(q, x)] = \theta$, which indicates θ' is an unbiased estimation of θ . $\square$

Based on the Lemmas 4 and 5, we have the following conclusion that reveals the correctness of the pruning condition:

LEMMA 6. Given a tolerant error $\delta \in (0, 1)$, z is the quantile of the standard normal distribution at $1 - \delta/2$. For two points q and x , denote θ' as the estimator of their angle under the m -length quantization, $\theta = \langle q, x \rangle$ falls into the confidential interval $[\theta' - z\sqrt{\theta'(\pi - \theta')/m}, \theta' + z\sqrt{\theta'(\pi - \theta')/m}]$ with the probability $1 - \delta$.

PROOF. As $\beta(q, x) \sim B(m, \theta/\pi)$, for the success probability $p = \theta/\pi$, the $1 - \delta$ confidence interval of p can be $[p' - z\sqrt{\frac{p'(1-p')}{m}}, p' + z\sqrt{\frac{p'(1-p')}{m}}]$ based on the Wald method [1], where $p' = \beta(q, x)/\pi$ is the proportions of successes. By rewriting it, the confidential interval of θ can be $[\theta' - z\sqrt{\theta'(\pi - \theta')/m}, \theta' + z\sqrt{\theta'(\pi - \theta')/m}]$. $\square$

Based on this lemma, $\theta \geq \theta' - z\sqrt{\theta'(\pi - \theta')/m}$ holds with the probability $1 - \delta$. When it holds, $I_{ub} = \|q\| \cdot \|x\| \cdot \cos(\theta' - z\sqrt{\theta'(\pi - \theta')/m}) \geq q^\top x = \|q\| \cdot \|x\| \cdot \cos \theta$. So, when I_{ub} is less than $I_{\min}$, it is proper to discard x .

6.2 Time and Space Complexity

To analyze the time and space complexity, we first discuss the setting of the related parameters in MARIA, including the maximum inner-block degree M , maximum inter-block degree between two blocks M' , the number of blocks s , the number of hash tables L and the number of hash functions used in a table K . As that in mainstream graph based methods [11, 35, 45, 45, 60], we fix M and M' as a constant. s depends on the range of norms in $\mathcal{D}$ and is

Table 2: Summary of Datasets

Datasets	Card.	Dim.	Size (GB)
YahooMusic	625,000	300	0.72
Deep1M	1,000,000	256	1.00
Tiny5M	5,000,000	384	4.77
Tiny80M	79,302,017	384	113
Deep1B	1,000,000,000	96	358
Yandex	1,000,000,000	200	745

affected little by n . Hence, $s = O(1)$. In addition, L and K can also be set as constants for a fast retrieval [25, 33].

The construction and query cost of MARIA depends on the cost of search in a block, C_Q . To construct APG, we conduct at most $O(\log s)$ searches in the APG for a point in $\mathcal{D}$. So the construction cost is $O(nC_Q \log s)$. Similarly, the worst query cost is $O(sC_Q)$ when all blocks are checked. The structure of an APG is so complicated that it becomes very difficult for analyzing C_Q [39, 51]. Some works [34, 50] regard C_Q as $O(\log n)$ by considering the “six degrees of separation” assumption [6, 54]. In addition, C_Q is proven to be $O(1)$ if the entry point is close enough to the query [56]. Based on these conclusions, C_Q is very likely to be bounded by $O(\log n)$. Consider that $s = O(1)$, the construction cost of MARIA is $O(n \log n)$ and the query cost in MARIA is $O(\log n)$. Then, each point has at most $O(M)$ inner-block edges and $O(M' + \log s)$ inter-block edges, so the index size is $O(n(M + M' + \log s)) = O(n)$.

THEOREM 1. *The index size of MARIA is $O(n)$. Assuming that the cost of search in an APG is $O(\log n)$, MARIA can be built in $O(n \log n)$ time and can answer a query within $O(\log n)$ cost.*

7 Experiment

In this section, we conduct extensive experiments on real-world datasets to provide a comprehension analysis of MARIA. We implement MARIA² and the competitors in C++ compiled with g++9.5.0 using -Ofast optimization and openMP for parallelism. All experiments are run on Ubuntu 22.04.5 LTS with 4 Intel(R) Xeon(R) Gold 6218 CPUs (160 threads) and 1.5 TB RAM.

7.1 Experimental Settings

Datasets. We employ 6 real-world datasets with varied cardinality of from 0.6M to 1B, whose information is summarized in Table 2 in the ascending order of their sizes. These datasets are widely used in ANNS and AMIPS [32, 35, 57]. Especially, the two billion-scale datasets, Deep1B and Yandex dataset³ are used for evaluating the scalability. For each dataset, we randomly select 10K query points from its query set to study the query performance.

Competitors. To demonstrate the indexing and query performance of MARIA, we compare it with the following AMIPS methods, including LSH-based and graph-based methods.

- **FARGO**⁴ [57] divides the dataset via norm partition where each block has at most N_0 points. Then, it adopts L K -dimensional hash tables to answer c -AMIPS problems in each block.

- **ip-NSW**⁵ [35] adopts the classic APG, HNSW [32] for solving AMIPS problem by replacing the similarity function in HNSW from $L2$ distance as *inner product*.
- **ip-NSW**⁺⁶ [32] improves the query performance of ip-NSW by building another APG in the angular space to generate the better entry points for search in ip-NSW.
- **NAPG**⁷ [45] improves the graph quality of ip-NSW by adjusting the edge selection strategy in ip-NSW for remaining more neighbors for each vertex.

Parameter Settings. We consider the k AMIPS queries with $k = 50$ for all algorithms in the default settings. Following the settings in the paper or source code of our competitors, we adopt the fixed parameters for each algorithm. For MARIA, we set $\alpha = 0.85$, $L = 4$, $K = 16$, $M = 24$, $efC = 80$ and $M' = 16$. For ip-NSW, ip-NSW⁺ and NAPG, as they are all improved on top of HNSW, we set $M = 24$ and $efC = 80$ for them. For FARGO, we set $N_0 = 4096$, $c = 0.8$, $L = 5$ and $K = 12$ as suggested in the paper.

Evaluation Metric. We compare the algorithms in terms of index size, indexing time, QPS (queries per second) and average query accuracy. QPS indicates the number of queries answered by an algorithm. We adopt recall to measure the query accuracy. Given a query point q , for a k AMIPS query, assume that the algorithm return the set $R = \{x_1, \dots, x_k\}$ and the exact k MIPS of q is $R^* = \{x_1^*, \dots, x_k^*\}$, then recall are defined as follows [57],

$$\text{Recall} = \frac{|R \cap R^*|}{k}. \quad (8)$$

7.2 Self Evaluation

In this subsection, we demonstrate the effectiveness of our three optimizations: the LSH framework, the skip-list-like mechanism, and the quantization-based pruning condition, on both the indexing and query performance of MARIA. We compare MARIA with the following variations: (i) **MARIA-noL**, which omits the LSH index; (ii) **MARIA-FC**, where points in a block $\mathcal{D}_i$ connect to all other blocks $\mathcal{D}_j$ for $j > i$; and (iii) **MARIA-noQ**, which excludes quantizations and pruning conditions. Due to space constraints, we report results only on the YahooMusic dataset, as shown in Figure 5.

7.2.1 Evaluation of the LSH Framework. We compare MARIA with MARIA-noL to demonstrate the effectiveness of the LSH framework. Regarding indexing performance, MARIA has a 10% larger index size due to the LSH framework’s space consumption but spends 25% less time constructing the index. This indicates that the higher-quality entry points generated by the LSH framework reduce the search cost in the APG. For query performance, MARIA achieves higher recall with slightly lower QPS compared to MARIA-noL, suggesting that MARIA-noL tends to terminate the search at suboptimal local maxima due to random entry points.

7.2.2 Evaluation of the Skip-List-like Mechanism. We compare MARIA with MARIA-FC to evaluate the skip-list-like mechanism. For indexing performance, MARIA-FC results in a 3–4× larger index size and 30% higher indexing time than MARIA, due to the increased number of inter-block edges. However, the increase in

²<https://github.com/SIGMOD25-MARIA/MARIA>

³<https://storage.yandexcloud.net/yandex-research/ann-datasets/T2I/base.1B.fbin>

⁴<https://github.com/Jacyhust/FARGO>

⁵<https://github.com/stanis-morozov/ip-nsw>

⁶<https://github.com/jerry-liujie/ip-nsw/tree/GraphMIPS>

⁷https://github.com/ThrunGroup/BanditMIPS/tree/camera_ready/algorithms/napg

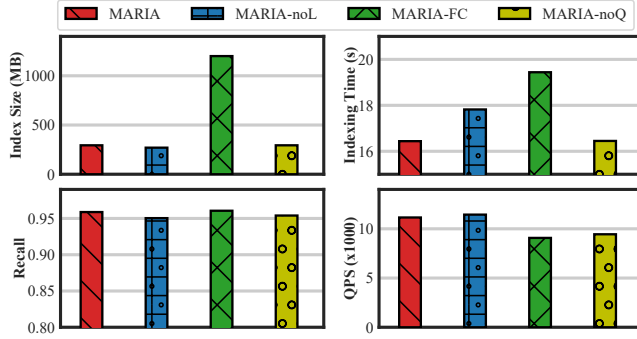


Figure 5: Self Evaluation

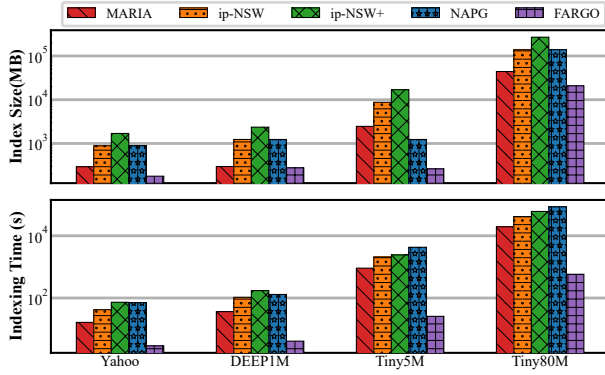


Figure 6: Indexing Performance

indexing time for MARIA-FC is less pronounced than the index size, indicating that finding inter-block edges incurs a lower cost than finding inner-block edges due to smaller candidate sets. For query performance, MARIA-FC achieves higher recall but significantly lower QPS, indicating that the skip-list-like mechanism maintains query accuracy while greatly reducing query cost.

7.2.3 Evaluation of the Quantization-Based Pruning. We compare MARIA with MARIA-noQ to assess the quantization-based pruning condition. For indexing performance, MARIA and MARIA-noQ exhibit similar index sizes and indexing times. However, regarding query performance, MARIA achieves higher QPS than MARIA-noQ while maintaining nearly the same recall, indicating that the pruning condition effectively filters out many unqualified points. Comparing MARIA-noL, MARIA-noQ, and MARIA, we find that the LSH framework enhances indexing performance more significantly, while quantization-based pruning improves query performance. This difference arises because the indexing phase prioritizes faster construction with lower query quality, making better entry points more beneficial. However, due to limited query accuracy during APG construction, the pruning condition is less frequently utilized.

7.3 Evaluation of Indexing Performance

In this set of experiments, we evaluated the index size and indexing time of all algorithms. Our findings are illustrated in Figure 6. We made the following observations when comparing all algorithms based on these two metrics. (1) Among the graph-based algorithms,

MARIA has the smallest index size, surpassed only by the LSH-based method, FARGO. This occurs because ip-NSW, ip-NSW⁺, and NAPG utilize multi-layer structures and store data points within the indexes, leading to significant space consumption. FARGO achieves the smallest index size by storing only $L = 5$ hash tables. Both ip-NSW and NAPG have similar index sizes due to their identical maximum degree, which determines the space allocated for the APG. Ip-NSW⁺ has the largest index size as it requires an additional APG to generate entry points. (2) Among the graph-based algorithms, MARIA has the smallest indexing time, approximately 50% of the second-best method, ip-NSW, highlighting the effectiveness of the LSH framework and quantization-based pruning condition in enhancing indexing efficiency. Ip-NSW⁺ or NAPG exhibits the largest indexing time on different datasets. The high construction cost of ip-NSW⁺ stems from the additional cost of building another APG. NAPG’s construction efficiency varies with datasets as it adjusts edge selection rules based on data distribution, leading to unstable construction costs. Additionally, the high construction cost of NAPG on Tiny5M and Tiny80M highlights the inefficiency of its proposed edge selection strategy.

7.4 Evaluation of Query Performance

In these experiments, we analyze the relationship between recall and QPS using the Recall-QPS curve to evaluate the query performance of all algorithms. The results for four datasets are shown in Figure 7. NAPG exhibits significantly poorer performance on Tiny5M and Tiny80M, so its results are not reported for these datasets. From the figures, we observe the following: (1) As QPS decreases, all algorithms achieve higher recall, consistent with the trade-off between recall and QPS. Among all algorithms, MARIA achieves the highest QPS for the same recall, indicating the best balance between query quality and efficiency. This superiority is due to MARIA’s improved graph structure, better entry points generated in LSH indexes, and an effective pruning condition for filtering out neighbors. (2) FARGO achieves the second-best query performance on YahooMusic, Tiny5M, and Tiny80M but surprisingly exhibits the lowest performance on DEEP1M. This difference arises from the varied norm distributions of these datasets. In YahooMusic, Tiny5M, and Tiny80M, the wide range of point norms allows finding satisfactory AMIPS results in a few blocks after norm partition. However, in DEEP1M, points have nearly identical norms. Since FARGO limits the maximum number of points in a block, the MIPS results are widely distributed across all blocks, complicating the search. In contrast, MARIA uses only the norm ratio α to control block size, maintaining efficiency on such datasets. The difference in FARGO’s query performance highlights the effectiveness of norm partitioning in the AMIPS problem. (3) The recall of ip-NSW and ip-NSW⁺ nearly converges at 0.95 on Tiny5M and 0.85 on Tiny80M, even when increasing the candidate size ef to boost query accuracy, indicating a significant norm bias issue. Combined with the analysis on FARGO, this underscores the critical importance of norm partitioning in the AMIPS problem.

7.5 Scalability Test

In this set of experiments, we evaluate the scalability of all algorithms on two billion-scale datasets, DEEP1B and Yandex. We

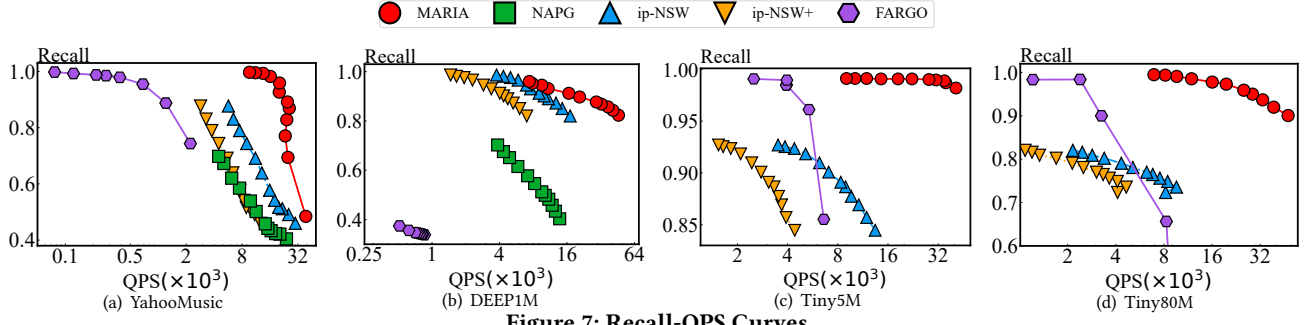


Figure 7: Recall-QPS Curves

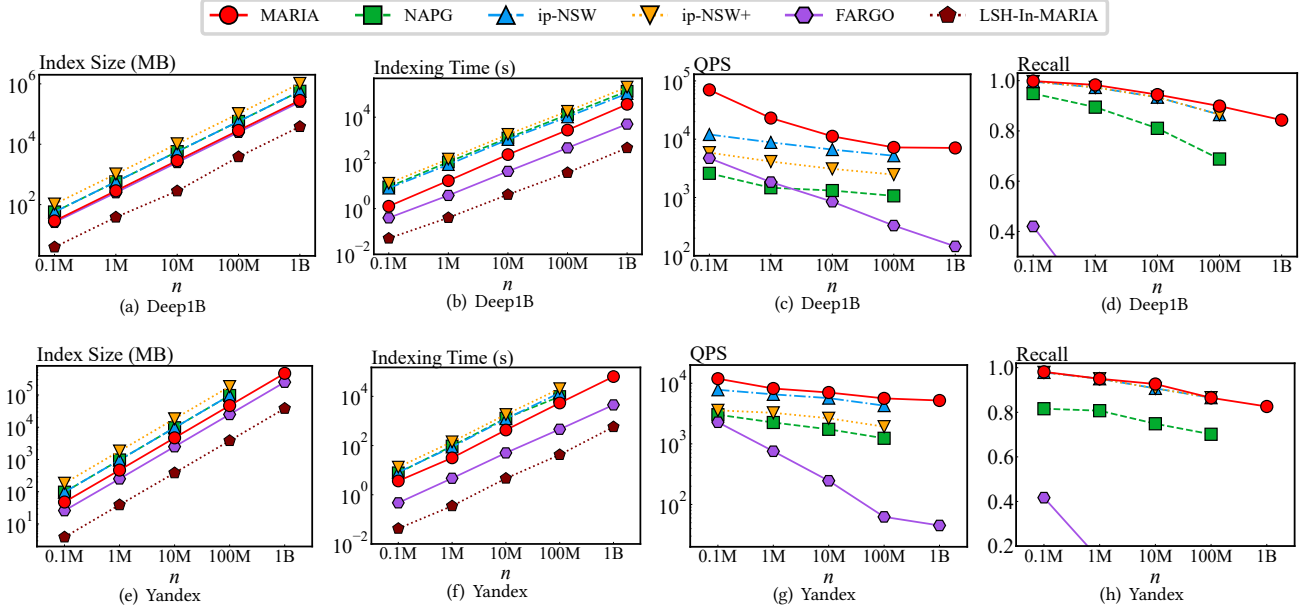


Figure 8: Scalability Test

assess indexing performance and query performance by varying data cardinality in the range of $\{0.1M, 1M, 10M, 100M, 1B\}$. Additionally, we examine the extra index size and indexing time from the LSH framework in MARIA (denoted as LSH-In-MARIA). ip-NSW, ip-NSW⁺ and NAPG fails to build indexes on Yandex due to out of memory, so we do not report their performance on Yandex. All results are shown in Figure 8. From the figure, we observe the following: (1) MARIA achieves the smallest index size and indexing time among all graph-based methods, indicating strong support for large-scale datasets. Moreover, the indexing time and index size of LSH-In-MARIA are less than 1% of those in MARIA, demonstrating its lightweight and efficiency. Additionally, as an LSH index, LSH-In-MARIA shows superior indexing performance compared to FARGO. (2) Comparing index sizes across different datasets, we found that index size is nearly linear with dataset cardinality n in all algorithms. This is because these algorithms have fixed degrees or a fixed number of hash tables. The indexing time also follows this trend, indicating that the cost of finding neighbors for a point changes slightly with data cardinality n . (3) MARIA consistently achieves the highest QPS and recall among all algorithms on these two datasets. Moreover, MARIA's QPS decreases very gradually

with n . Notably, when n changes from 100M to 1B, MARIA's QPS remains nearly unchanged, further demonstrating its superior scalability in query performance. (4) Although FARGO achieves a much smaller index size and indexing time than all graph-based methods, its query performance is poor, especially for datasets with large cardinality. As n increases, FARGO's QPS decreases rapidly, becoming uncompetitive with graph-based methods. FARGO's inferior QPS is due to its $O(n)$ theoretical query cost. The query accuracy of FARGO also dramatically decreases with n , highlighting the limited query performance of LSH-based methods on large-scale datasets.

8 Conclusions

In this paper, we introduce a novel framework called MARIA for AMIPS queries in high-dimensional spaces on large-scale datasets. MARIA addresses challenges such as the norm bias issue and the non-metric nature of the inner product by using a norm partition-based graph structure. We also implement three optimizations to enhance query performance and scalability. Extensive experiments show that MARIA achieves a 2-4 $\times$ speedup in indexing and query costs compared to leading competitors, while providing superior query quality, making it suitable for billion-scale datasets.

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